



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Mid-Term Exam: Trimester: Summer 2025

Course Code: EEE 2123; Course Title: Electronics
Total Marks: 30; Duration: 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Q1. Consider a diode is operating at $(40+x)^{\circ}\text{C}$. [x = final digit of your ID]

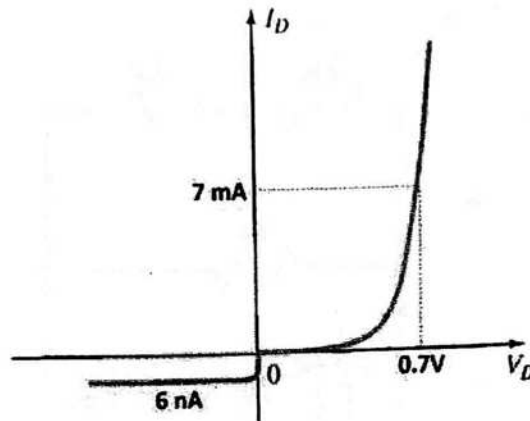


Fig 1: Circuit Diagram for Q1

- (a) Find the ideality factor (n) of the diode. [2] ✓
- (b) Qualitatively draw the IV curves at 20°C , 80°C compared to $(40+x)^{\circ}\text{C}$ in a single plot and mark the temperatures. [2] ✓
- (c) Calculate the static resistances when i) 0.8V ii) -0.8V is applied across the diode. [2]

Q2. Determine the values of $I_1, I_2, I_3, I_4, I_5, V_1, V_2, V_3, V_4$ and V_5 from the circuit of Fig. 2 [6]

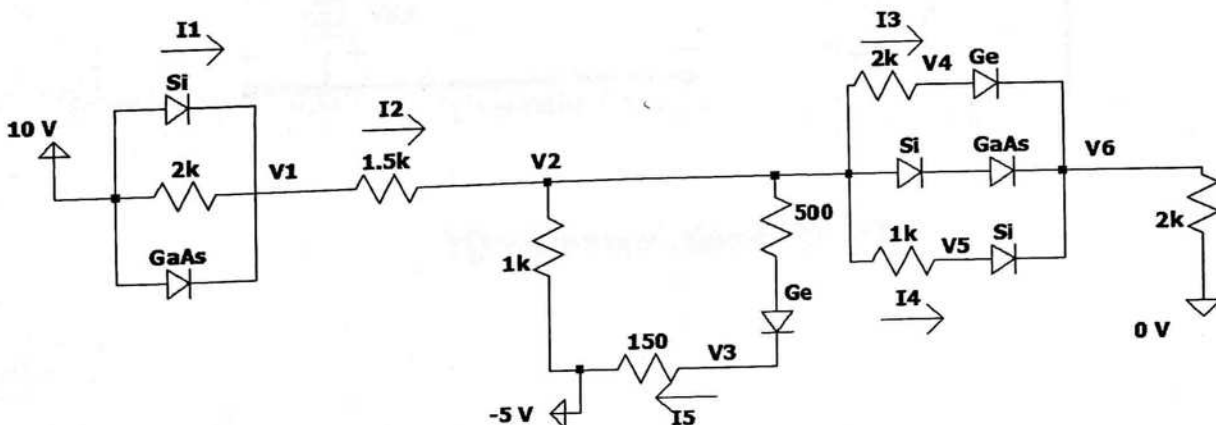
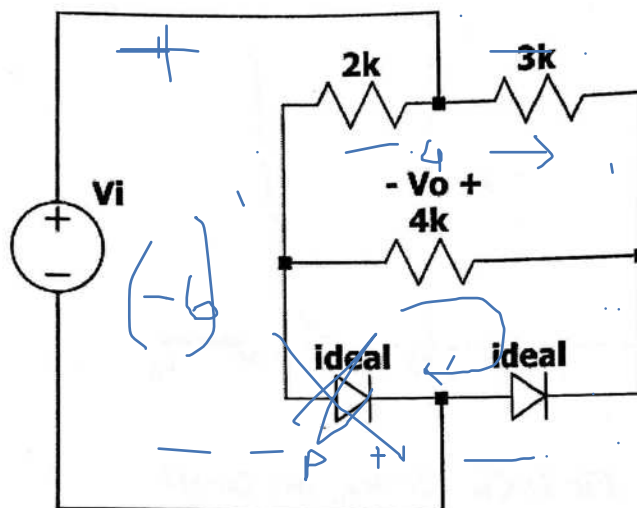
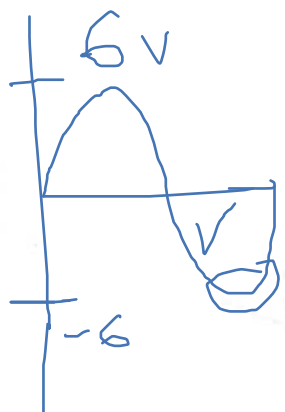


Fig 2: Circuit diagram for Q2

Q3. The input to the circuit of Fig 3 is a sine wave with an amplitude of 6 V. The diodes have a breakdown voltage of 4 V. [4]

(a) Find the expression of output voltage and sketch it. [2]

(b) Explain mathematically whether the circuit will be safe or not.



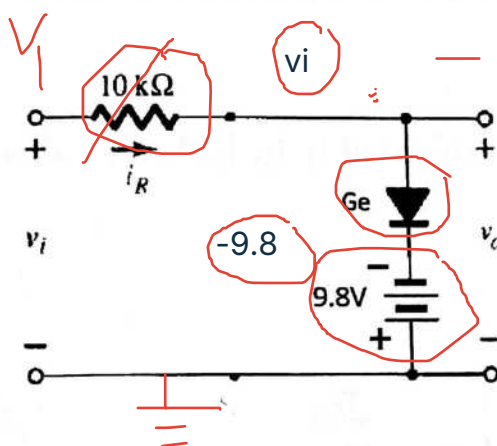
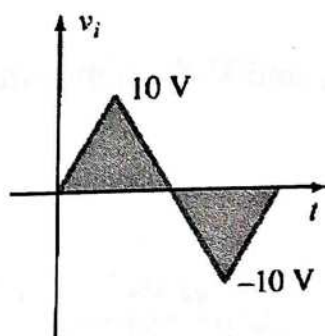
$$+P - 4 = 0$$

$$P = 4$$

$PIV < br = \text{safe}$
 $piv \geq br ; \text{unsafe}$

Fig 3: Circuit diagram for Q3

Q4. Analyze and sketch V_o and find i_r for the network of the following circuit for the input shown below. Clearly mention the peak values in your sketch. [3+3]



Insert text here

Fig 4: Circuit diagram for Q4

diode on

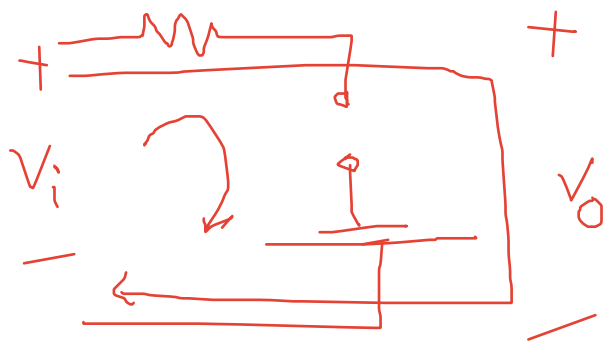
$$v_i + 9.8 > .3$$

$$v_i > -9.8 + 0.3$$

$$v_i > -9.5$$

V_P

$V_n > V_K$

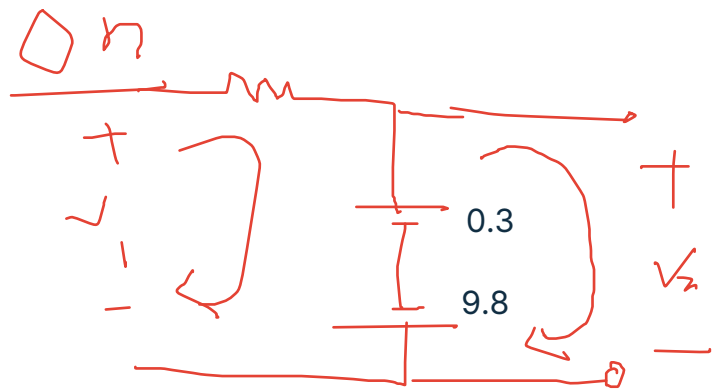
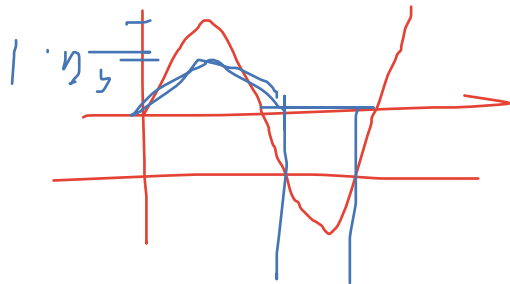


$$i = 0$$

$$-v_i + i r + v_o = 0$$

$$v_o = v_i$$

$$v_i < -9.5$$



$$v_o = -(9.8 - 0.3) = -9.5$$

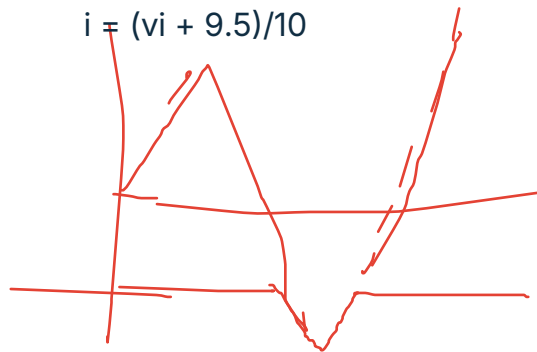
$$v_i > -9.5$$

$$+9.8 - 0.3 + v_o = 0$$

$$v_o = -9.8 + 0.3 = -9.5$$

$$-v_i + 10i + 0.3 - 9.8 = 0$$

$$i = (v_i + 9.5)/10$$



Q5. Consider the following diagram,

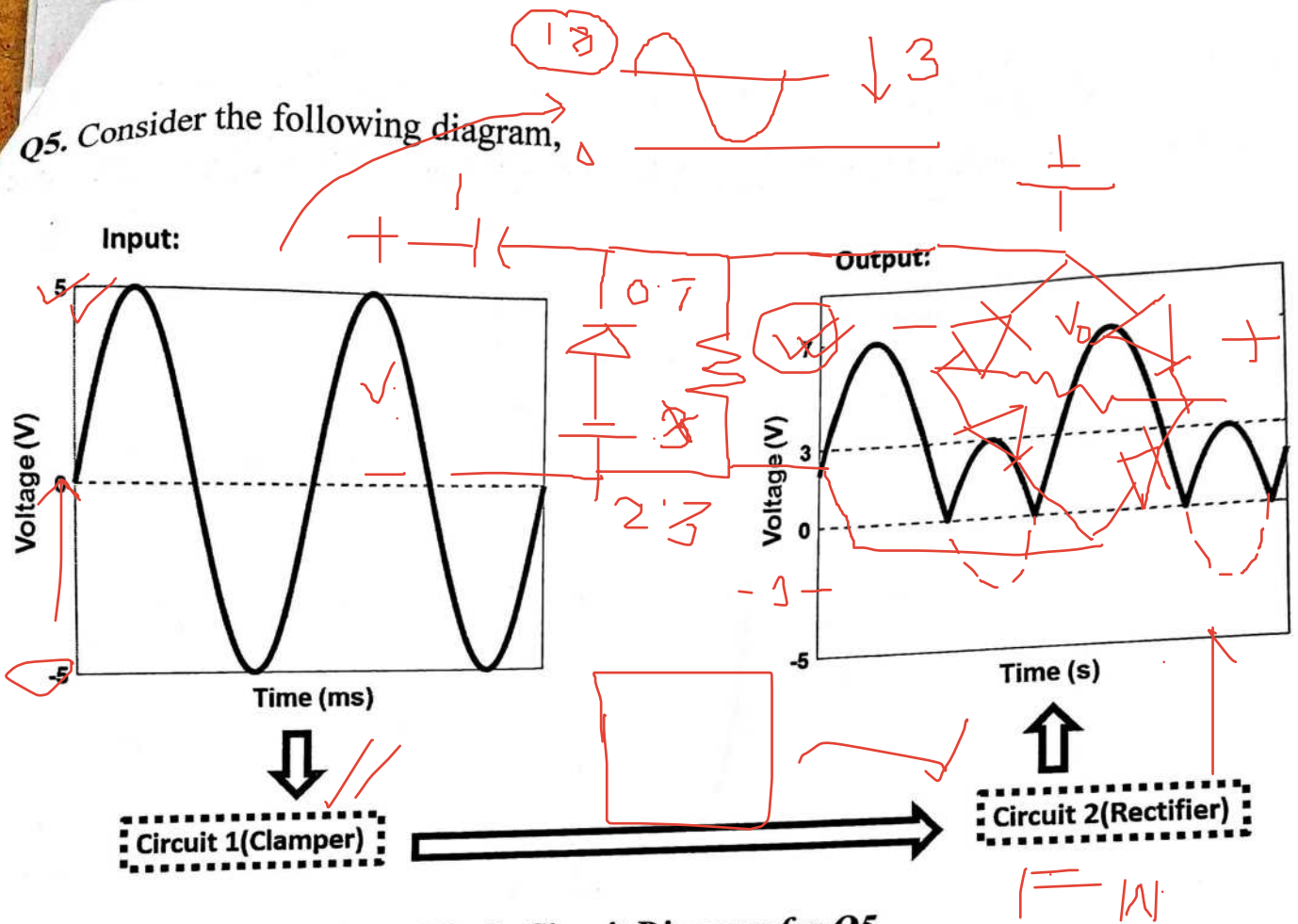
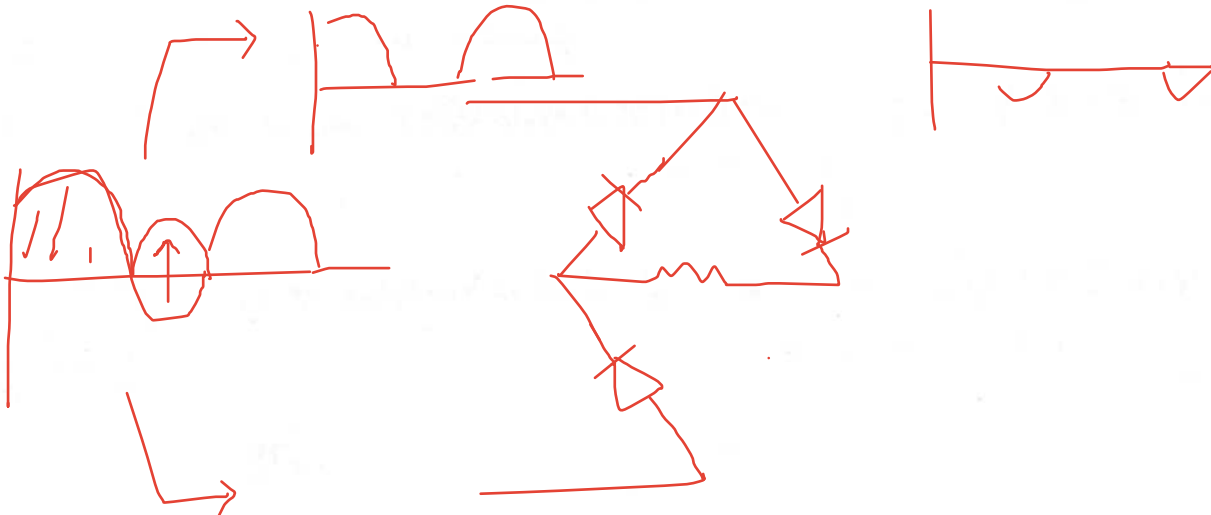


Fig 5: Circuit Diagram for Q5

The given input waveform is first modified by **Circuit 1 (Clamper)** and then further processed by **Circuit 2 (Rectifier)** resulting in the final output waveform. Design **Circuit 1** using Si diodes and **Circuit 2** using ideal diodes. [3+3]





United International University

Name
(Optional)

ID No.

Section

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Signature with date

Course Code

Trimester / Semester: Spring / Summer / Fall, 20.....

Date:

Name of Exam: Class Test / Mid-term / Mid-term (Makeup/Improvement) / Final / Final (Makeup/Improvement)

Q.1

(a)

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$

$$\Rightarrow \left(\ln \left(\frac{I_D}{I_S} \right) + 1 \right) = \frac{V_D}{nV_T}$$

$$\Rightarrow n = \frac{V_D}{V_T \left[\ln \left(\frac{I_D}{I_S} \right) + 1 \right]}$$

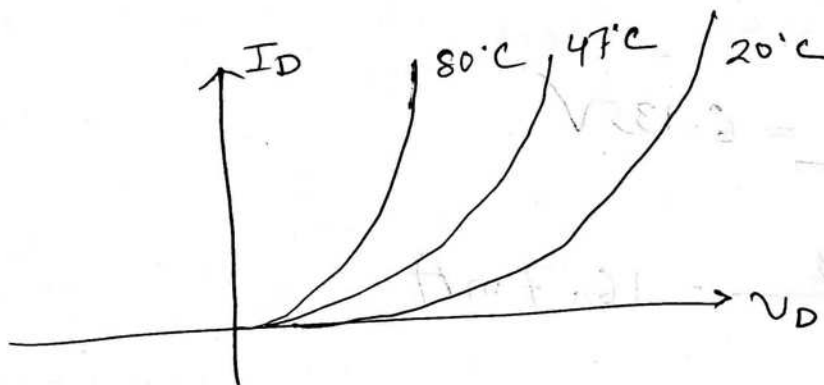
$$= \frac{0.7}{0.0276 \left(\ln \left(\frac{7 \times 10^{-3}}{6 \times 10^{-9}} \right) + 1 \right)} = 1.69$$

$$T = 47^\circ\text{C}$$

$$V_T = \frac{k \times (273 + 47)}{q}$$

$$= 0.0276$$

(b)



$$(c) \quad I_D = 6 \times 10^{-9} e^{\frac{0.8}{nV_T}}$$

$$= 0.1685 \text{ A}$$

$$R_{(0.8V)} = \frac{0.8}{I_D} = 4.74 \Omega$$

$$\text{at } V_D = -0.8V, \quad I_D = -6 \text{ nA}$$

$$\therefore R_{(-0.8V)} = \frac{-0.8}{-6 \times 10^{-9}} = 133.33 \text{ m}\Omega$$

$$\boxed{8.2} \quad V_1 = 10 - 0.7 = 9.3V$$

Applying KVL from +10 V to 0V segment,

$$-10 + 0.7 + 1.5I_2 + 0.7 + 1.2 + 2I_2 = 0$$

$$\Rightarrow I_2 = 2.11 \text{ mA}$$

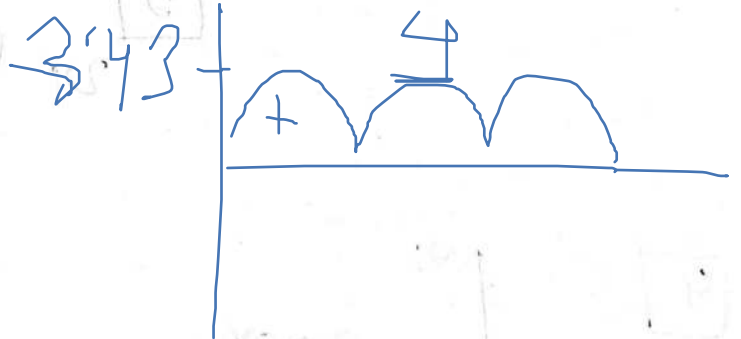
$$V_2 = V_1 - 1.5I_2 = 6.135V$$

$$I_5 = \frac{V_2 + 5 - 0.3}{0.65} = 16.7 \text{ mA}$$

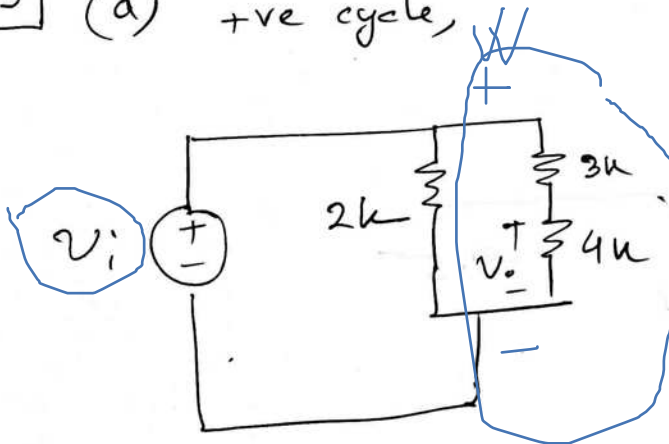
$$I_3 = \frac{0.7 + 1.2 - 0.3}{2} = 0.8 \text{ mA}$$

$$I_4 = \frac{0.7 + 1.2 - 0.7}{1} = 1.2 \text{ mA}$$

$$V_6 = V_2 - 0.7 - 1.2 = 4.235 \text{ V}$$



[3] (a) +ve cycle,



$$v_o = \frac{4}{7} v_i = \frac{4}{7} \times 6 = 3.43$$

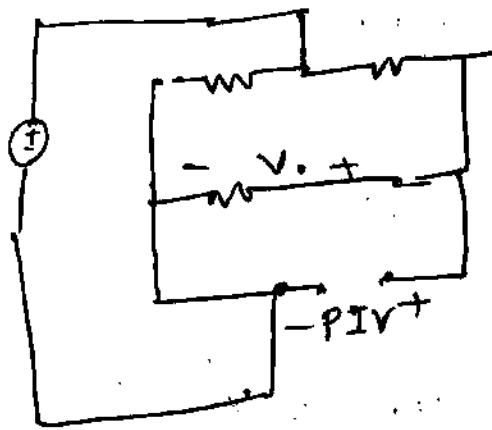
-ve cycle

$$v_o = \frac{4}{6} \times |v_i| = 4 \text{ V}$$



$$\frac{4}{7+2} V_i$$

(8)



$$-PIV + V_o = 0$$

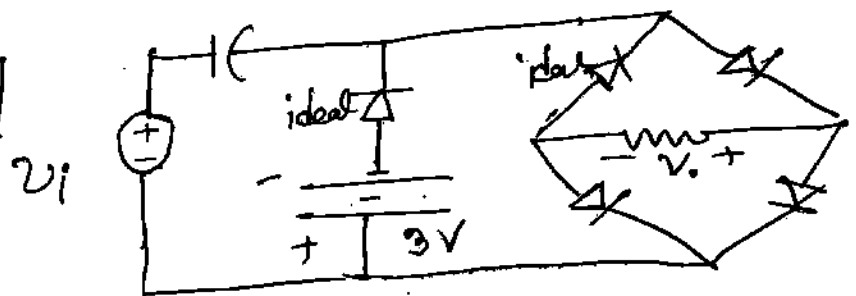
$$\Rightarrow PIV = V_o = 3.43V$$

at negative cycle

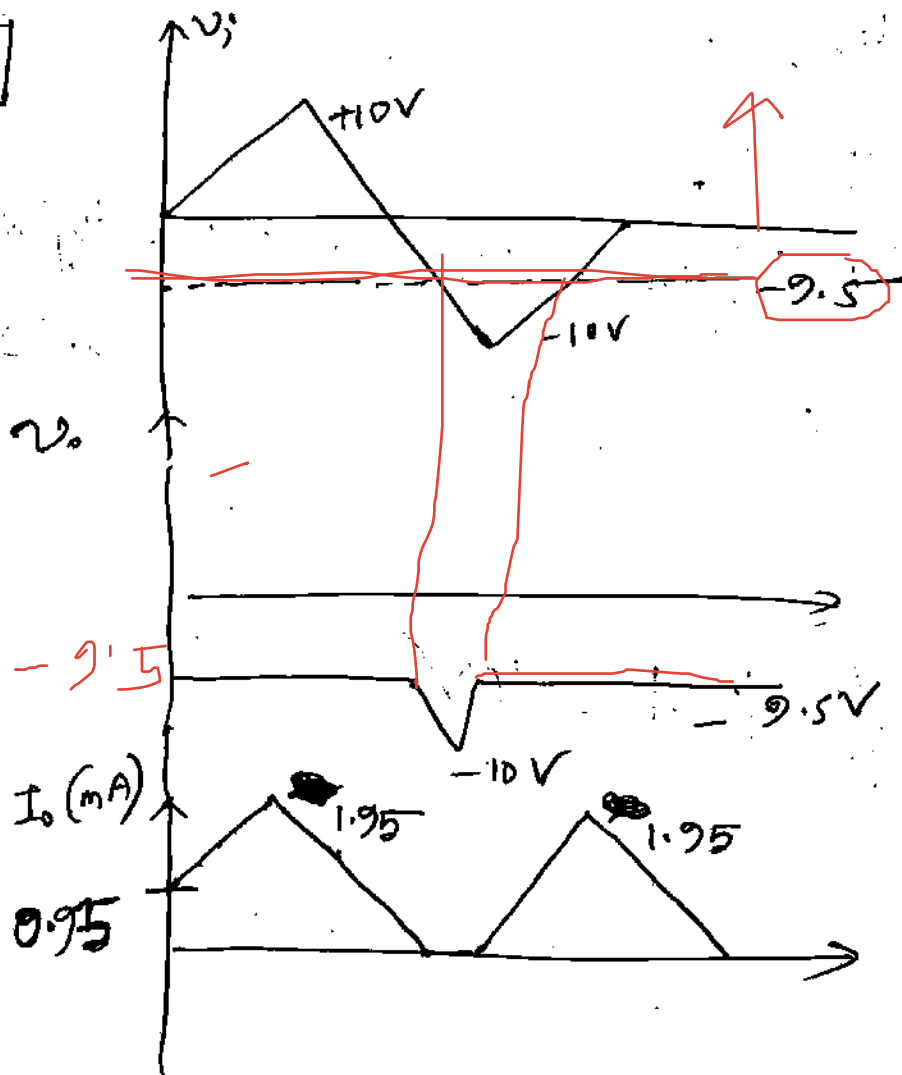
$$PIV = 4V$$

So not safe.

[5]



[4]



$$i = \frac{V_i - V_o}{10} \text{ mA}$$